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324871 (24)

BE (8th Semester)

Examination, April-May, 2014

Branch : Electrical, EEE

EHV AC&DC TRANSMISSION (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory and carry two marks. Attempt any two parts from (b), (c) and (d). Each question of (b), (c) and (d) carry 7 marks.

UNIT - I

Q. 1. (a) Name the different types of H.V.D.C. systems. 2

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(b) What are the advantages and limitations o H.V.D.C. systems ?

(c) Draw a schematic diagram of an E.H.V. A transmission line and explain the need f intermediate substations.

(d) Explain the commutation and overlap ang for a 6-pulse rectifier bridge.

UNIT - II

Q. 2. (a) Define a double circuit line in reference to long AC transmission line.

(b) Name different types of static VAR syste and explain thyristor controlled rectifier (TC fixed capacitor type (FC) systems.

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(c) Explain with neat sketch the voltage profile along a long EHV AC line with

(i) light load and with

(ii) heavy load 7

(d) What is a flexible AC transmission ? Why is it necessary ? 7

UNIT - III

3. (a) Name the different types of lightning arresters used in EHV transmission systems. 2

(b) A surge of 150 KV travels on a line of surge impedance 450 ohms towards the junction of the line with two branch lines. The surge

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impedance of branch lines are 400 ohms an 40 ohms. Find the surge voltage and curre transmitted into each branch line. Also fin the reflected surge voltage and current. 4

(c) A step wave of 200 KV travels on a line havin surge impedance of 500 ohms and reache the end of the line where the line is terminate by an inductance of 2500 μH. Find the voltag across the inductance and the reflecte voltage wave. 7

(d) What type of lightning stroke can occur o EHV transmission systems ? How the

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overvoltages can be controlled which arise due to lightning strokes. 7

UNIT - IV

Q. 4. (a) Define high - pass filter. 2

(b) (i) On what factors is the reactive to power requirement of a converter station depend ?

(ii) Explain in brief as to why inverter end requires higher reactive power supply than the rectifier end ? 7

(c) What are the harmful effects of harmonics caused by operation of HVDC converter ?

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What remedial measures are taken suppress them ?

(d) Compare between MTDC and interconnection and explain advantages MTDC systems.

UNIT - V

Q. 5. (a) Define valve - blocking.

(b) Explain by means of a schematic diagra and equations, how power flow through a H.V.D.C. link is controlled.

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(c) Write short notes on any one of the following :

(i) Constant ignition angle control

(ii) Constant extinction angle control 7

(d) Describe parallel operation of H.V.D.C. link with an A.C. Network. 7

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